



DoctorApp

SPECIFICATION & WALKTHROUGH

Core Application Features & Navigation Structure

A comprehensive report documenting the React Native Doctor Booking Application's architecture, user interface elements, functional flows, and layout navigation stack.

PREPARED BY

Antigravity Coding Assistant

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PROJECT SCOPE

Day 4 Deliverable

BUILD ENVIRONMENT

React Native v0.85.3 (Android)

1. Executive Summary & App Overview

DoctorApp is a premium, cross-platform mobile application built using React Native. The system is designed to bridge the gap between patients and healthcare professionals by offering an elegant, secure, and crash-free appointment booking flow. Emphasizing a modern flat design, intuitive grid selectors, and strict data validation, the application delivers a premium healthcare scheduling experience.

Key Design Goals

The application was structured around three key pillars: high visual fidelity (using curated color palettes and modern typography), bulletproof routing (resetting navigation stacks where appropriate to prevent state mismatch), and complete offline-first simulated data flows.

2. Core Application Screens & Features

Below is a breakdown of the primary screens implemented in DoctorApp, detailing their structural elements, controls, and dynamic behaviors.



Splash Screen & Onboarding

Serves as the entry point of the app, establishing the brand and app icon visual context immediately on launch.

- Custom-designed digital branding logo displaying a medical cross with pulse line.
- Automatic 2-second timeout transition redirecting to the Authentication flow.
- Styled StatusBar integration providing a premium status background match.



Authentication Screens (Login / Register)

Secures client sessions with input verification, validation warnings, and automatic flow redirects.

- **Mobile & Password Verification** : Enforces a strict 10-digit mobile check and non-empty password verification.
- **Visual Feedback** : Dynamic red outline borders and error label blocks highlight problematic fields.
- **Registration Redirect** : Seamless transition between Login and Register with automated success banner routing.



Dashboard & Doctor Search

The centralized hub for selecting doctors, featuring full search filters and category tags.

- **Search Bar** : Instant search filtering across doctor names, specialties, and clinics.
- **Specialty Chips** : Quick filtering based on categories (Dentists, Cardiologists, Pediatricians, etc.).
- **Doctor Profiles Cards** : Cards listing experience, ratings, clinic name, fees, and next available times.



Doctor Details (About / Reviews Tab)

Presents deep profile information using an elegant state-based tab view switcher.

- **Tab Switcher** : Stateless toggles to swap between "About" and "Reviews" views.
- **Reviews Feed** : Renders a list of client reviews showing dates, comments, and star ratings (★/☆).
- **Booking Call-to-Action** : Footer button directing users to the scheduling flow.



7-Day Slot Picker Selector

A dynamic date-and-time selector containing slot availability verification.

- **Horizontal Date Picker** : Dynamically generates the upcoming 7 calendar days.
- **Availability Grid** : Grid showcasing time slots with custom states: White (Available), Blue (Selected), and Gray Strike-through (Booked/Disabled).
- **Validation Block** : Disable "Confirm Slot" button until a valid available slot is chosen.



Booking Summary & Patient Details

Consolidates scheduling, pricing, and visit reason metadata before final checkout.

- **Doctor Summary Card** : Displays selected clinician overview.
- **Fee Breakdown** : Detailed list showing Consultation Fee, Platform Fee, and Total Cost.
- **Patient Info & Reason Box** : Captures patient contact credentials and optional visit reasons.



Booking Confirmation & Navigation Control

A completion state providing validation of success, Booking ID generation, and navigation stack resetting.

- **Randomized Booking ID** : Auto-generates a unique identifier (e.g. `BK104928`).
- **Stack Purge** : Clicking "Go to Home" resets the navigation stack, preventing back navigation into booking states.

3. Application Navigation & Screen Flow

DoctorApp uses the **React Navigation Native Stack** framework to govern transitions. All navigations utilize strict type parameter mapping, routing values from the Dashboard down to the final Confirmation page.

Splash Screen



Triggers auto-redirect after 2 seconds

Login Screen



Performs input validation checks

Dashboard (Home)



Doctor lists search and category filters

Doctor Profile



Toggles between About/Reviews, links to scheduling

Slot Picker



Selects active date and available time slot

Booking Summary



Review schedules, reasons, and pricing lists

Confirmation



Displays randomized BK ID; resets navigation to Home

4. Screen Navigation Stack Reference

Below is the technical reference directory for all screens registered in the main stack navigator (`AuthNavigator.js`).

Screen Name	Configuration Title	Required Route Parameters	Transition Reset
Splash	None (Hidden)	None	YES (TO LOGIN)
Login	"Login"	<code>`successMessage`</code> (Optional)	YES (TO HOME)
Register	"Register"	None	No (Standard Back)
Home	"Find Your Doctor"	None	No (Stack Root)
DoctorProfile	"Doctor Details"	<code>`doctor`</code> (Object)	No (Standard Back)
SlotPicker	"Select Slot"	<code>`doctor`</code> (Object)	No (Standard Back)
BookingSummary	"Booking Summary"	<code>`doctor`</code> , <code>`selectedDate`</code> , <code>`selectedSlot`</code>	No (Standard Back)
BookingConfirmation	None (Hidden)	<code>`doctor`</code> , <code>`selectedDate`</code> , <code>`selectedSlot`</code> , <code>`bookingId`</code> , <code>`patientName`</code>	YES (TO HOME)

5. Technical Specifications & Architecture

The app is developed with high-quality structure patterns, featuring modular layout files, clean separation of concerns, and local state isolation.

Key Technical Patterns

- **Parametric Navigation** : Rather than relying on global state configurations, components pass transaction states (`doctor`, `date`, `slot`) down the navigation pipeline using safe React Navigation `route.params`.
- **Deterministic Scheduling UI** : Uses the JavaScript native `Date` API to calculate and render the upcoming 7 calendar days relative to the client's current time.
- **AAPT Resource Compliance** : Launch resources and app icons have been strictly compiled to high-density PNG file signatures, resolving Android asset compilation errors.
- **Style Sheet Separation** : Inline styles are omitted. All layout rules are strictly structured into dedicated React Native `StyleSheet` modules with clean hierarchy.